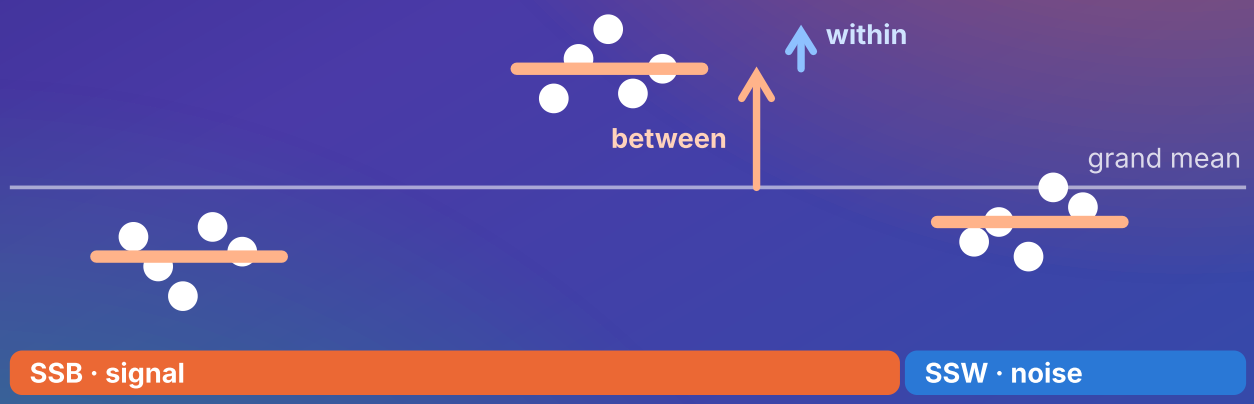
A WORKED TEXTBOOK CHAPTER · SECOND EDITION

ANOVA from First Principles

Variance decomposition, regression, signal versus noise, and the survey cluster design effect, told as one story.



$$SST = SSB + SSW$$

15 chapters · 15 figures · 12 applications · 32 graded exercises · Tested R code · Survey DEFF & margin of error · Punjab-style survey lab

Part I Foundations · Part II Applications · Part III Surveys · Part IV Practice

2026

How to use this chapter

This chapter treats ANOVA not as "a test for three or more means" but as an **accounting system for variation**. Every idea (the F test, regression, interactions, the intraclass correlation and the survey design effect) comes from the same single identity: *total variation = variation we can attribute to structure + variation we cannot*.

Colour code used everywhere · Total variation (SST) · Signal: between groups (SSB) · Noise: within groups (SSW)

PART I · FOUNDATIONS

1 Why ANOVA exists
2 Variance from first principles
3 The ANOVA decomposition
4 The F statistic and hypothesis test
5 What is signal and what is noise?

PART II · APPLICATIONS

6 Twelve one-way applications, and what comes after F
7 ANOVA inside regression: farm yield
8 Two-way ANOVA and interactions

PART III · SURVEYS

9 Clustering, ICC and the design effect (DEFF)
10 Survey lab: a Punjab-style pre-poll design

PART IV · PRACTICE

11 Assumptions, diagnostics, and a decision guide
12 Formula sheet
13 Do it in R (and Python)
14 Exercises: 32 questions, easy → hard
15 Answer key

By the end you should be able to…

Decompose
Split any outcome's variation into between-group and within-group parts, by hand and in R.

Test & size
Run and read an F test, then report *how big* the effect is (η^2, ω^2, Tukey intervals).

Diagnose
Check independence, equal spread and normality; switch to Welch or mixed models when needed.

Connect
See regression, two-way ANOVA and random effects as the same decomposition.

Plan surveys
Turn an ICC into a DEFF, an effective sample size and an honest margin of error.

Optimise
Choose how many interviews per polling booth give the most precision for a fixed budget.

WHAT IS NEW IN THIS EDITION

- **New figures:** variance drawn as squares; one observation split into three deviations; the F distribution; a 2 × 2 "signal vs noise" panel; a gallery of all twelve examples; Tukey intervals; residual diagnostics; interaction plots; a simulation of SRS vs cluster sampling; the cost-optimal cluster size.
- **Two new "reality-check" examples:** a genuine *null* result ($F \approx 0.7$), and an unequal-variance case where classical ANOVA and Welch's ANOVA disagree sharply.
- **Statistical corrections:** the farm regression's predictors are almost perfectly collinear ($r = 0.996$, VIF ≈ 123), so neither slope is individually significant; this is now explained. Its coefficients and F are recomputed from the printed data. The two-way example now shows *all* of its data so it can be reproduced. The design-effect formula is described as exact only under its model. ω^2 is added because η^2 overstates effects in small samples.
- **New material:** a proof that the cross-product vanishes; what F's numerator estimates; post-hoc tests; DEFT and margins of error; a Punjab-style worked survey design; the optimal number of interviews per booth; a tested R companion script; 7 new exercises (26–32).

PART I

Foundations

variance · sums of squares · degrees of freedom · the F ratio · signal vs noise

1 Why ANOVA exists

Suppose you observe different outcomes across people, farms, schools, machines, constituencies, hospitals, or survey clusters. The interesting question is never simply *whether* observations differ (they always do). The useful question is:

How much of the observed variation lines up with some systematic structure, and how much is left over?

Analysis of variance (ANOVA) answers by splitting a total sum of squared deviations into pieces. In the simplest case it compares how far apart the **group means** are with how scattered the **observations inside each group** are. If the means are far apart *relative to* the scatter inside groups, the grouping variable carries information about the outcome.

Total variation
How much do all observations spread around the grand mean?

=

Explained (signal)
Spread that lines up with groups, predictors or clusters

+

Residual (noise)
Spread left over within groups, around the fitted model

The same algebra underlies ordinary least-squares regression: a categorical factor is just a set of 0/1 indicator variables. Multiple regression, factorial ANOVA, ANCOVA, mixed models and multilevel survey models are all extensions of this one principle, and Part III shows that the survey **design effect** comes from it too.

2 Variance from first principles

2.1 Why deviations must be squared

For observations $y_1, \ldots, y_n$ with mean $\bar{y}$, the raw deviations $y_i - \bar{y}$ always add to exactly zero: the positives cancel the negatives. Squaring them stops the cancelling and gives extra weight to large departures. The resulting total is the **total sum of squares**:

$$\mathrm{SST} = \sum_{i=1}^{n} (y_i - \bar{y})^2, \qquad s^2 = \frac{\mathrm{SST}}{n-1}.$$

You can take "squared deviation" literally: each one is the *area of a square* whose side is the distance from the mean.

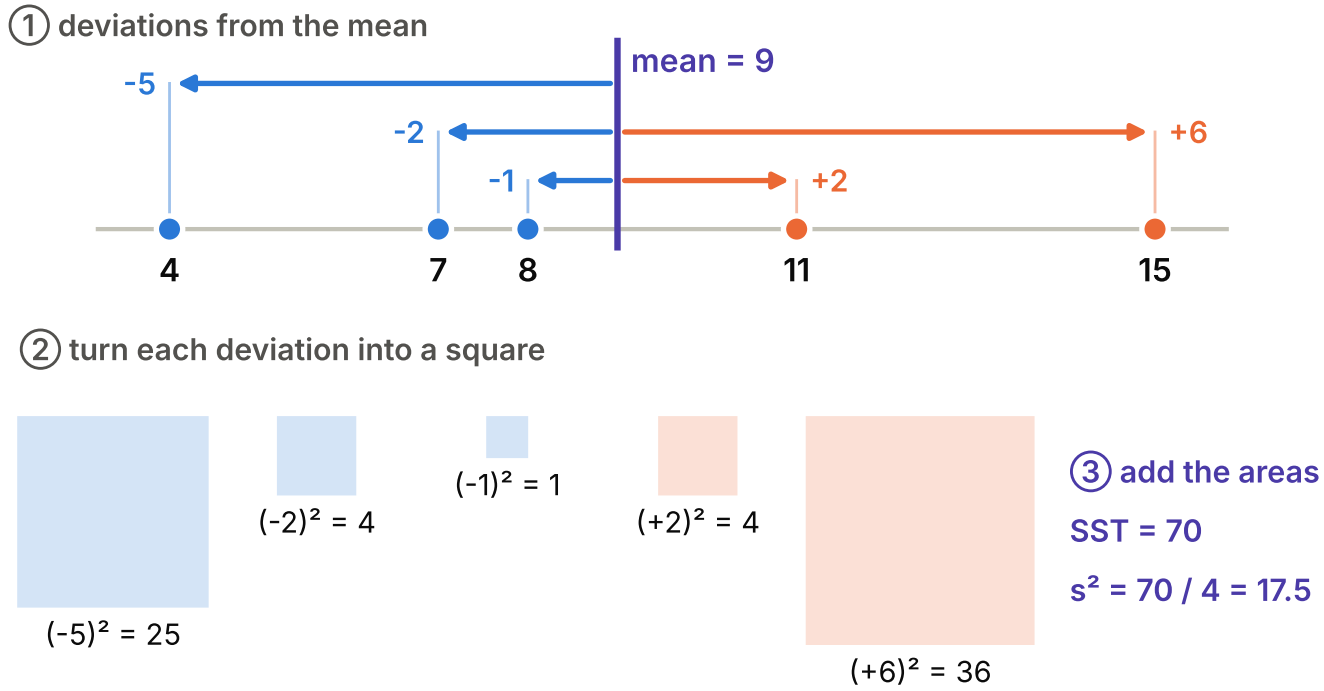


Figure 1. Variance is literally an average area. Five values (4, 7, 8, 11, 15) with mean 9. Top: each value's deviation from the mean (blue below, orange above). Bottom: each deviation turned into a square whose side is that distance, so its area is the squared deviation. Points far from the mean (15) contribute far more than points near it (8).

2.2 A tiny example

y	y − ȳ	(y − ȳ)²
8	−2	4
10	0	0
12	+2	4
ȳ = 10	**sum = 0**	**SST = 8**

The mean is 10, SST = 8, so the sample variance is $8/(3-1) = 4$ and the standard deviation is 2.

Notice the middle column sums to zero; this is always true, and it is exactly why we square.

2.3 Why divide by n − 1?

Once $\bar{y}$ has been computed from the data, only $n-1$ of the deviations are free to vary: the last one is forced so that they all sum to zero. These free pieces are the **degrees of freedom**. Dividing by $n-1$ rather than n makes s^2 an unbiased estimate of the population variance σ^2. The same bookkeeping, "sum of squares ÷ its degrees of freedom", produces every *mean square* in ANOVA.

3 The ANOVA decomposition

Let y_{ij} be observation j in group i ($i = 1, \ldots, k$; group i has n_i units; $n = \sum n_i$). Write $\bar{y}_i$ for the group mean and $\bar{y}$ for the grand mean. Add and subtract the group mean:

$$\underbrace{y_{ij} - \bar{y}}_{\text{total}} = \underbrace{(\bar{y}_i - \bar{y})}_{\text{between}} + \underbrace{(y_{ij} - \bar{y}_i)}_{\text{within}}$$

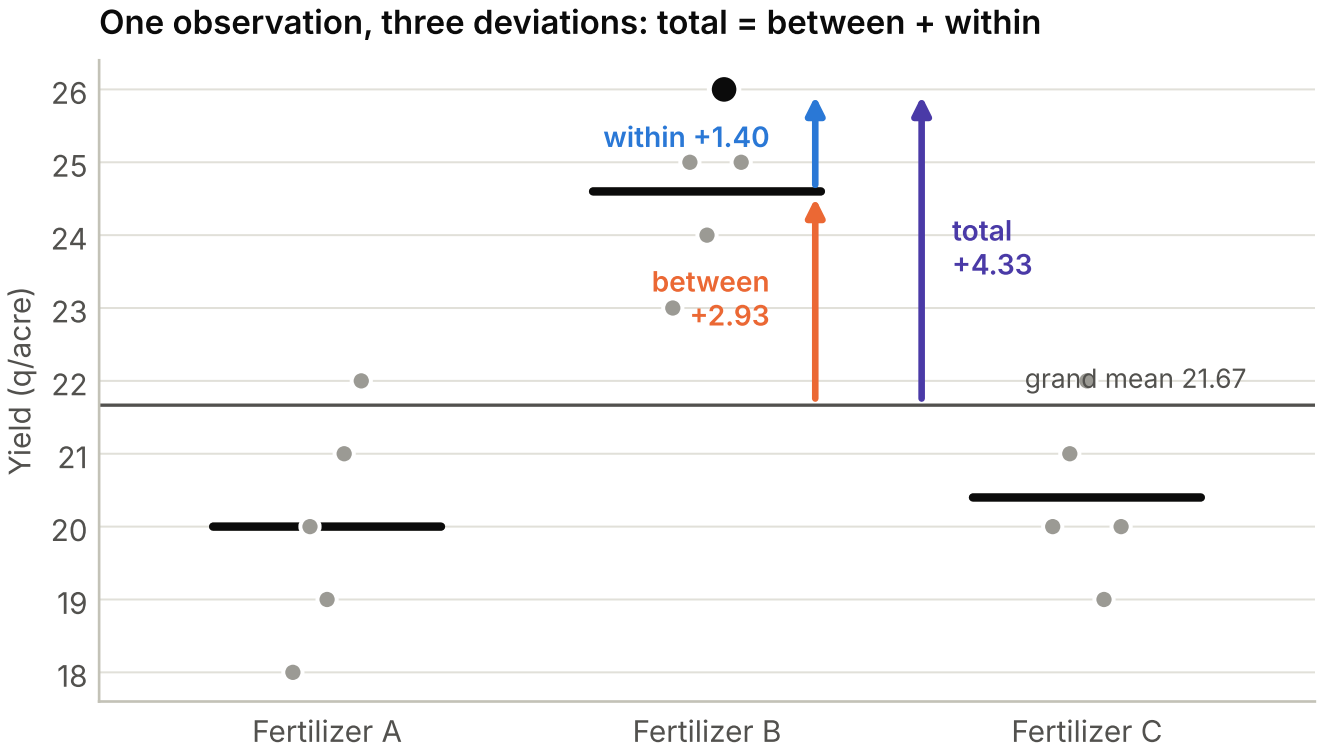


Figure 2. The decomposition, for one farm. The highlighted plot under Fertilizer B yielded 26 q/acre. It sits +4.33 above the grand mean (violet). That total splits into +2.93 because fertilizer B does better than average (orange, between) and +1.40 because this plot beat its own group mean (blue, within). Every one of the 15 points can be split the same way.

Square both sides and sum over every observation. The cross-product term disappears, leaving the **fundamental identity**:

$$\text{SST} = \text{SSB} + \text{SSW}, \qquad \text{SSB} = \sum_i n_i(\bar{y}_i - \bar{y})^2, \quad \text{SSW} = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2.$$

✓ WHY THE CROSS-PRODUCT VANISHES (THE STEP MOST BOOKS SKIP)

Expanding the square gives $\text{SST} = \text{SSB} + \text{SSW} + 2\sum_i \sum_j (\bar{y}_i - \bar{y})(y_{ij} - \bar{y}_i)$. For a fixed group i, the factor $(\bar{y}_i - \bar{y})$ is a constant, so it comes out of the inner sum: $2\sum_i (\bar{y}_i - \bar{y}) \left[\sum_j (y_{ij} - \bar{y}_i)\right]$. The bracket is the sum of deviations of group i from *its own* mean, which is always zero (Section 2.1). So the whole cross term is zero, for any data. ∎

3.1 Degrees of freedom and mean squares

Piece	Sum of squares	Degrees of freedom	Mean square	Why that df?
Between	SSB	k − 1	MSB = SSB/(k − 1)	k group means, minus 1 for the grand mean
Within	SSW	n − k	MSW = SSW/(n − k)	n observations, minus 1 per group mean
Total	**SST**	**n − 1**	**s^2 = SST/(n − 1)**	**the df add up too: (k − 1) + (n − k)**

MS_W is especially important: under the classical model it estimates the common error variance σ^2, the "background scatter" left once each group is allowed its own mean.

3.2 Worked by hand: fertilizer and wheat yield

Fifteen plots, three fertilizers, five plots each (yield in quintals per acre).

Group	Observations	Mean	Var
A	18, 20, 19, 21, 22	20.00	2.50
B	23, 25, 24, 26, 25	24.60	1.30
C	20, 21, 22, 19, 20	20.40	1.30
All	**n = 15**	**21.67**	**6.10**

SSB $= 5[(20 - 21.67)^2 + (24.6 - 21.67)^2 + (20.4 - 21.67)^2] = 5(2.78 + 8.60 + 1.60) =$ **64.93**

SSW $= 4(2.5) + 4(1.3) + 4(1.3) =$ **20.40** (each group: $(n_i - 1)$ × its variance)

SST $= 64.93 + 20.40 =$ **85.33** ✓ (check: 14 × 6.10)

Source	SS	df	MS	F	p
Between groups	64.93	2	32.47	19.10	< 0.001
Within groups	20.40	12	1.70		
Total	**85.33**	**14**			

4 The F statistic and the hypothesis test

The one-way null hypothesis is $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$; the alternative is that at least one population mean differs. The test statistic is a ratio of mean squares:

$$F = \frac{\text{MS}_B}{\text{MS}_W} = \frac{\text{SSB}/(k-1)}{\text{SSW}/(n-k)} \sim F_{k-1,\, n-k} \text{ under } H_0.$$

Why should F be near 1 when H_0 is true? Look at what each mean square estimates on average:

$$E[\text{MS}_W] = \sigma^2, \qquad E[\text{MS}_B] = \sigma^2 + \frac{\sum_i n_i(\mu_i - \bar{\mu})^2}{k-1}.$$

When all μ_i are equal the extra term is zero, so numerator and denominator estimate the same σ^2 and their ratio hovers around 1. Real differences between means inflate only the numerator, pushing F upward. *F is therefore literally (noise + signal) ÷ noise.*

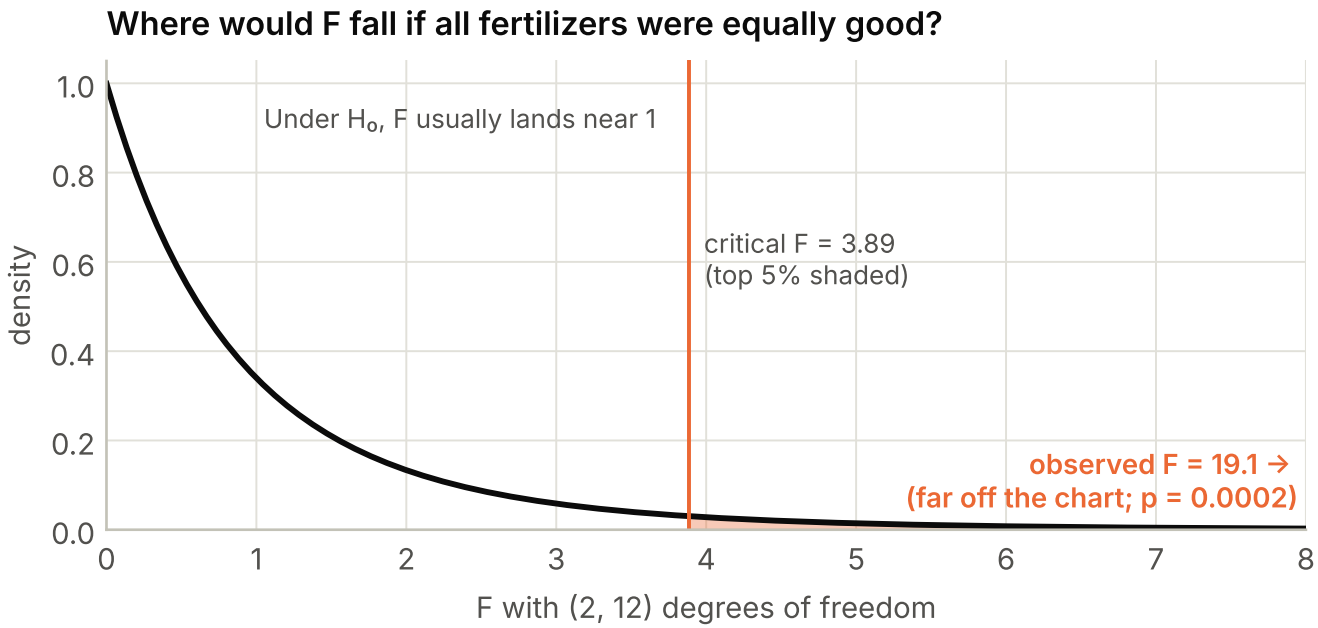


Figure 3. The reference distribution. If all three fertilizers were equally good, F would follow the F(2, 12) curve: usually near 1, and above 3.89 only 5% of the time (orange). The observed F = 19.1 is far beyond that, so data like these would be very surprising under H_0.

The **p-value** is the probability, assuming H_0 and the model's assumptions, of an F at least as large as the one observed. For the fertilizer data $p = 0.00019$.

⚠ COMMON MISTAKES WITH P-VALUES

- A small p says the means are *not all equal*. It does not say *which* differ (Section 6.3) or *by how much* (Section 6.2).
- A large p is *not* proof that the means are equal; it may just mean the study was too small or too noisy (Example 11).
- p is not the probability that H_0 is true.

5 What is signal and what is noise?

This is the conceptual heart of ANOVA. In a statistical model, **signal** means variation represented by the factors or predictors you specified. **Noise** means the residual variation left after fitting them. But "residual" does not automatically mean "meaningless randomness". It can contain measurement error, genuine individual differences, omitted predictors, curvature, interactions, time trends, and pure chance.

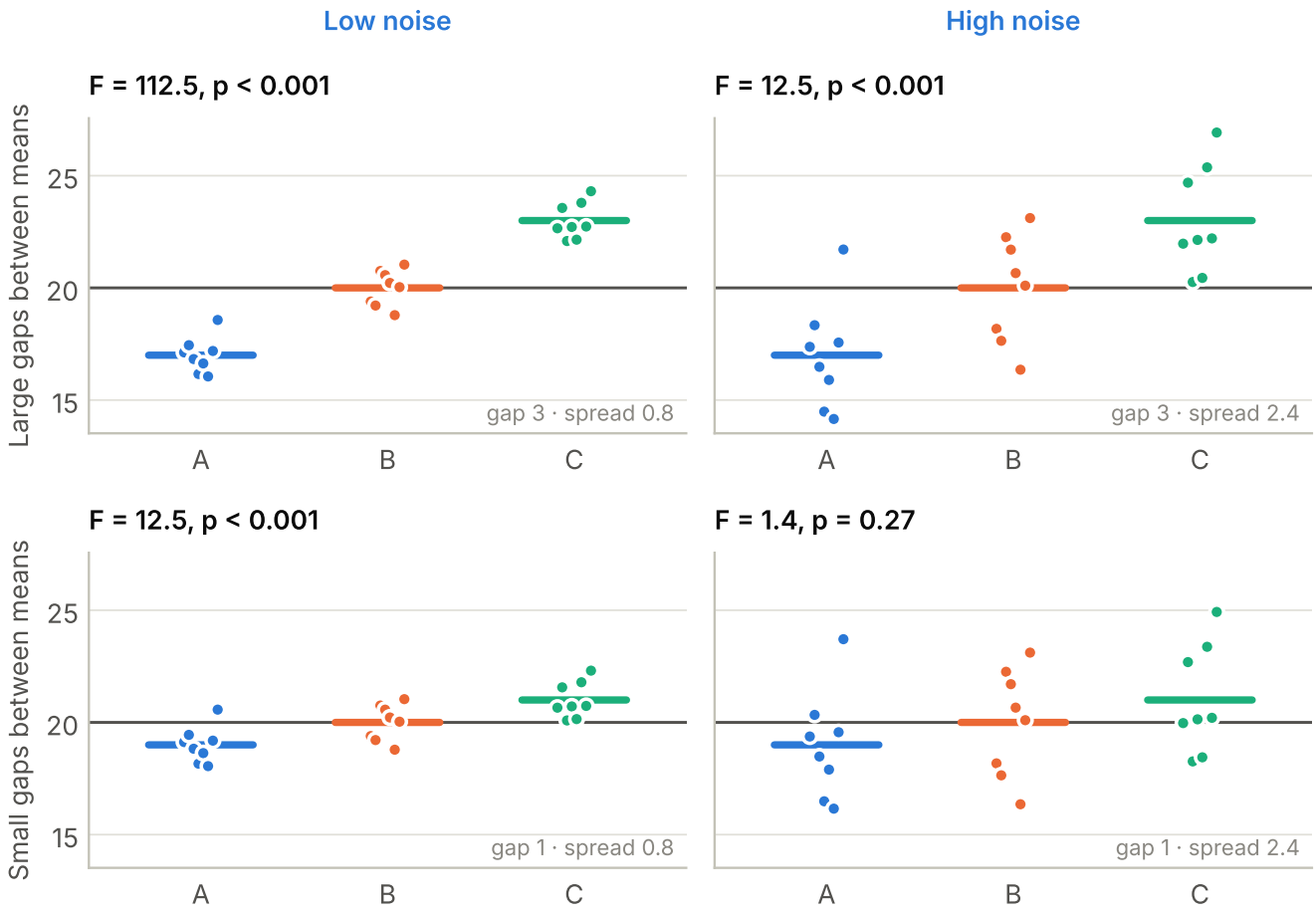


Figure 4. F compares the gaps between means (signal) with the scatter inside groups (noise). All four panels use the same shapes of scatter. Moving right triples the within-group spread; moving down shrinks the gaps between means. The top-right panel has *the same means* as the top-left, yet F collapses ninefold because the noise has tripled. And the top-right and bottom-left panels have *identical* F: what matters is the ratio of gap to spread, not either one alone.

5.1 How ANOVA estimates noise

Units that receive the same treatment give a natural estimate of background variability: their differences from their own group mean *cannot* be explained by the group factor. So SSW and MS_W estimate the error variance, provided the model is correct and within-group observations are independent with similar variance.

5.2 We cannot label an individual residual as "noise"

No calculation can look at one residual and certify that it is noise. The separation is statistical and depends on the model. Confidence that the residual variance really is noise comes from good *design* and *checking*:

- **Replication**
  Repeated comparable units measure background scatter.
- **Randomisation**
  Breaks links between treatment and hidden causes.
- **Blocking**
  Moves predictable variation (field, day, district) out of the residual.
- **Multilevel models**
  Separate person, household, village and PSU variance.

Residual plots
Reveal trends, funnels, outliers or curvature left behind.

Validation
Does the signal predict *new* data, or just this sample?

Reliability data
Repeat measurements show how much is instrument noise.

Intervals & power
Show how precisely the signal is pinned down.

PAUSE AND THINK

In a pre-election survey, respondents' vote intentions differ within a single polling booth. List three things, other than "randomness", that could be hiding in that within-booth variation.
(Ideas: age, caste or community, gender, household, interviewer, time of day.)

PART II

Applications

12 toy datasets | effect sizes | Tukey | Welch | regression | interactions

6 Twelve one-way applications, and what comes after F

Ten classic toy examples from agriculture, education, industry, marketing, health and energy, plus NEW two reality checks (highlighted). Each dataset has 3–4 groups of five observations. Look at the pictures before the numbers.

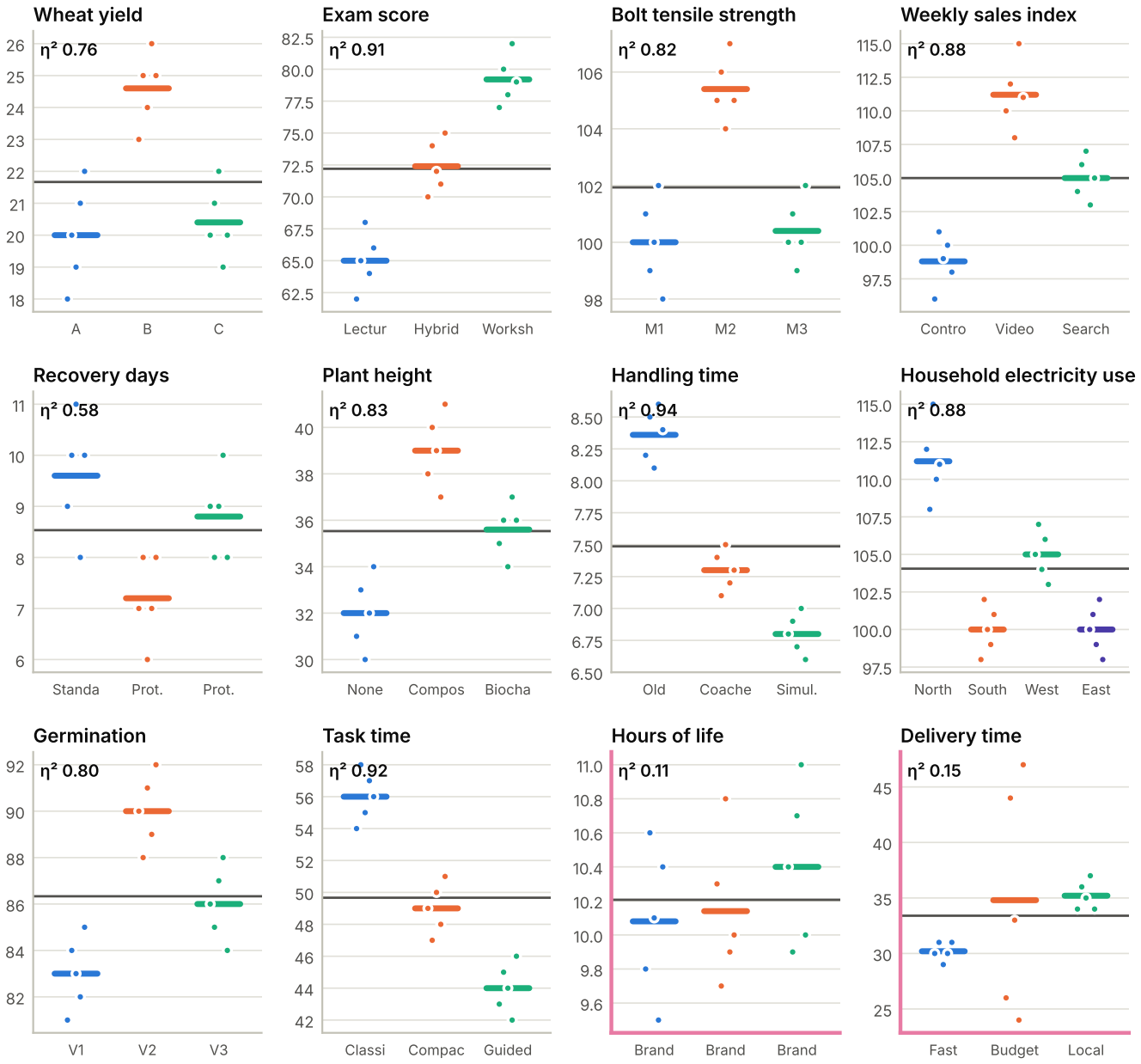


Figure 5. All twelve datasets at a glance. Dots are observations; short bars are group means; the grey line is the grand mean; η^2 is shown top-left. In examples 1–10 the groups barely overlap, which is why every one of them is "significant". The two pink-axis panels are the new reality checks: *battery life* (the means are practically on top of each other) and *delivery time* (one courier is wildly more variable than the others).

#	Application	Total var	MS_{within}	F	p	η^2	ω^2
1	Fertilizer type → wheat yield	6.10	1.70	19.10	< 0.001	0.761	0.707
2	Teaching method → exam score	39.74	4.33	58.20	< 0.001	0.907	0.884
3	Machine → bolt tensile strength	7.92	1.70	26.63	< 0.001	0.816	0.774
4	Ad campaign → weekly sales index	31.14	4.30	44.70	< 0.001	0.882	0.854
5	Treatment protocol → recovery days	1.84	0.90	8.30	0.005	0.580	0.493
6	Soil treatment → plant height	10.55	2.10	29.17	< 0.001	0.829	0.790
7	Call-centre training → handling time	0.48	0.031	102.34	< 0.001	0.945	0.931
8	Region → household electricity use	25.31	3.55	39.83	< 0.001	0.882	0.853
9	Seed variety → germination	10.95	2.50	24.67	< 0.001	0.804	0.759
10	Website layout → task time	28.10	2.50	72.67	< 0.001	0.924	0.905
11	Battery brand → hours of life	0.19	0.20	0.73	0.502	0.108	0.000
12	Courier → delivery time	36.97	36.70	1.05	0.379	0.149	0.007

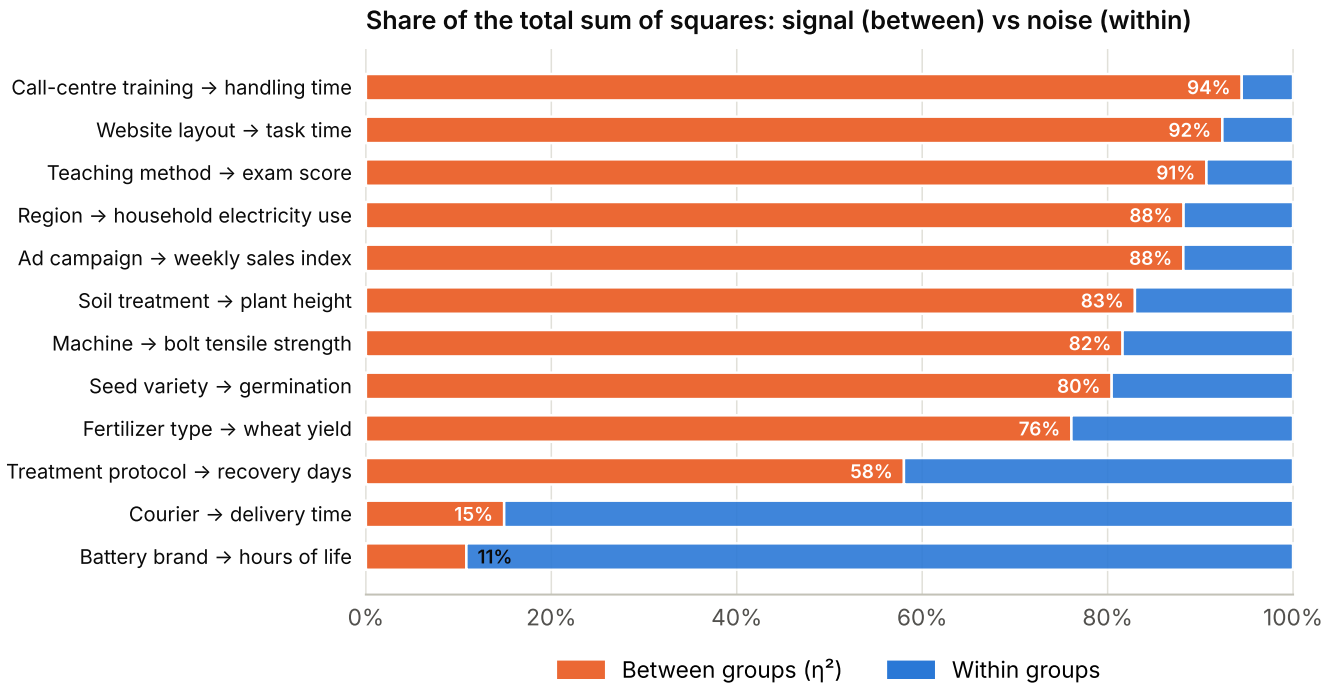


Figure 6. The "SS budget" of each example. Each bar is 100% of the total sum of squares, split into the share explained by the grouping (orange = η^2) and the share left within groups (blue). Sorted from least to most explained.

The raw data for all twelve examples are listed in the table on the next page, so that every row above can be reproduced by hand or with `aov()`.

#	Application	Groups and toy observations (group mean)
1	Fertilizer type → wheat yield unit: q/acre	**A**: 18, 20, 19, 21, 22 (20.00) **B**: 23, 25, 24, 26, 25 (24.60) **C**: 20, 21, 22, 19, 20 (20.40)
2	Teaching method → exam score unit: score	**Lecture**: 62, 65, 68, 64, 66 (65.00) **Hybrid**: 70, 74, 72, 71, 75 (72.40) **Workshop**: 77, 80, 78, 82, 79 (79.20)
3	Machine → bolt tensile strength unit: strength	**M1**: 101, 99, 100, 102, 98 (100.00) **M2**: 105, 106, 104, 107, 105 (105.40) **M3**: 100, 101, 99, 100, 102 (100.40)
4	Ad campaign → weekly sales index unit: index	**Control**: 96, 101, 99, 100, 98 (98.80) **Video**: 110, 112, 108, 115, 111 (111.20) **Search**: 104, 106, 107, 103, 105 (105.00)
5	Treatment protocol → recovery days unit: days	**Standard**: 9, 10, 11, 8, 10 (9.60) **Prot. B**: 7, 8, 6, 7, 8 (7.20) **Prot. C**: 9, 8, 9, 10, 8 (8.80)
6	Soil treatment → plant height unit: cm	**None**: 31, 33, 30, 32, 34 (32.00) **Compost**: 38, 40, 39, 37, 41 (39.00) **Biochar**: 35, 36, 34, 37, 36 (35.60)
7	Call-centre training → handling time unit: minutes	**Old**: 8.2, 8.5, 8.1, 8.6, 8.4 (8.36) **Coached**: 7.4, 7.1, 7.5, 7.2, 7.3 (7.30) **Simul.**: 6.8, 6.9, 6.7, 7, 6.6 (6.80)
8	Region → household electricity use unit: index	**North**: 112, 108, 115, 110, 111 (111.20) **South**: 98, 102, 100, 99, 101 (100.00) **West**: 105, 107, 104, 106, 103 (105.00) **East**: 100, 101, 99, 102, 98 (100.00)
9	Seed variety → germination unit: %	**V1**: 81, 84, 83, 82, 85 (83.00) **V2**: 90, 88, 91, 89, 92 (90.00) **V3**: 86, 85, 87, 88, 84 (86.00)
10	Website layout → task time unit: seconds	**Classic**: 54, 58, 55, 57, 56 (56.00) **Compact**: 49, 47, 50, 48, 51 (49.00) **Guided**: 43, 45, 44, 42, 46 (44.00)
11	Battery brand → hours of life unit: hours	**Brand X**: 9.8, 10.6, 10.1, 9.5, 10.4 (10.08) **Brand Y**: 10.3, 9.7, 10.8, 9.9, 10 (10.14) **Brand Z**: 10.4, 9.9, 10.7, 11, 10 (10.40)
12	Courier → delivery time unit: hours	**Fast**: 30, 31, 29, 30, 31 (30.20) **Budget**: 26, 44, 33, 24, 47 (34.80) **Local**: 34, 36, 35, 37, 34 (35.20)

6.1 Reading the summary table

"Total var" is SST/(n − 1), the variance you would compute ignoring groups. MS_{within} is what remains after each group gets its own mean. The ratio between them is a quick sense-check. For the call-centre data the total variance of 0.48 shrinks to 0.03 once training type is known; for the batteries it barely changes (0.19 → 0.20).

6.2 How big is the effect? η^2 and ω^2

$$\eta^2 = \frac{\text{SSB}}{\text{SST}} \qquad \omega^2 = \frac{\text{SSB} - (k-1)\,\text{MS}_W}{\text{SST} + \text{MS}_W}$$

η^2 is the share of the sample's SST associated with the grouping. It is *biased upward*: even pure noise produces some SSB. ω^2 subtracts the SSB we would expect from noise alone, so it estimates the population share more honestly. For fertilizer: $\eta^2 = 0.761$ but $\omega^2 = 0.707$. With only five observations per group the gap matters. For batteries $\eta^2 = 0.108$ while ω^2 is negative (reported as 0).

6.3 Which groups differ? Post-hoc comparisons

A significant omnibus F says *at least one* mean differs. To find which, compare pairs while controlling the chance of any false alarm across all comparisons. Tukey's Honest Significant Difference (HSD) is the standard choice for all pairwise comparisons.

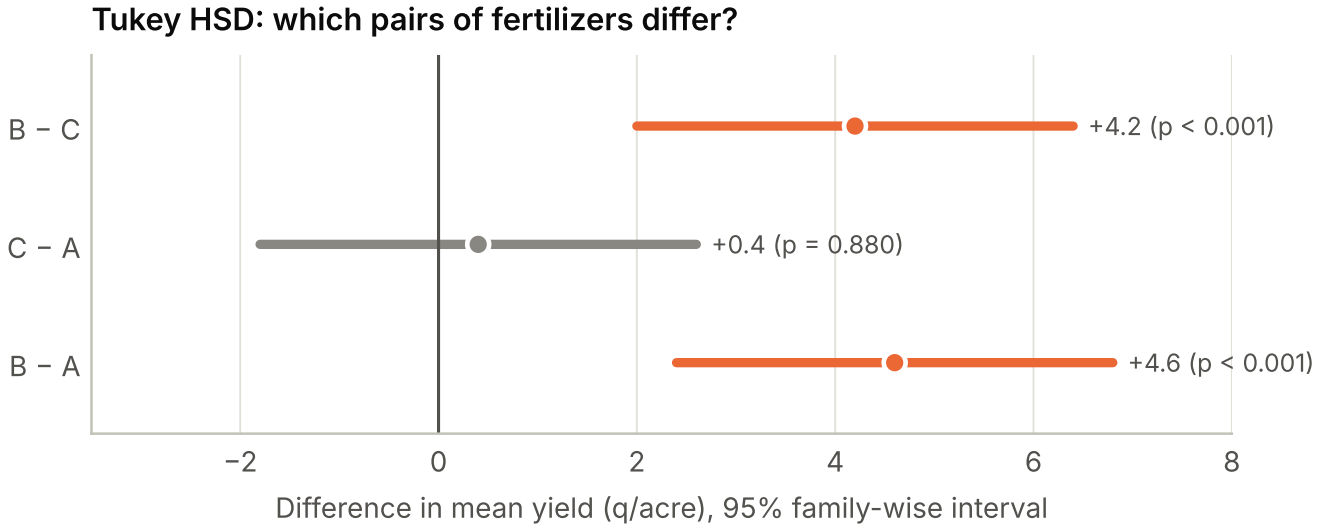


Figure 7. Tukey 95% family-wise intervals for the fertilizer data. B beats both A and C by about 4–5 q/acre (orange intervals exclude zero). A and C cannot be told apart (grey interval straddles zero). This is the practical conclusion the F test alone could not give.

6.4 NEW Example 11: a null result (battery life)

Three brands, five batteries each: means 10.08, 10.14 and 10.40 hours. F = 0.73, p = 0.50. F is below 1, which is exactly what equal means would typically produce.

Source	SS	df	MS	F	p
Between brands	0.29	2	0.14	0.73	0.502
Within brands	2.38	12	0.20		
Total	**2.67**	**14**			

⚠ ABSENCE OF EVIDENCE ≠ EVIDENCE OF ABSENCE

It is tempting to write "the brands are identical". But with n = 5 per brand, a 95% interval for any pairwise difference is roughly ±0.61 hours wide. The honest statement is: *"We found no evidence of a difference; the data are consistent with differences of up to about 0.6 hours either way."* Report the interval, not just the p-value.

6.5 NEW Example 12: unequal spreads (courier delivery time)

The Fast and Local couriers are extremely consistent (variances 0.7 and 1.7), while Budget is erratic (variance 107.7). Classical ANOVA pools all three variances into one $MS_{within} = 36.7$, which is dominated by Budget. That inflated "noise" swamps the clear 5-hour gap between Fast (30.2 h) and Local (35.2 h).

F = 1.05, p = 0.38
Classical one-way ANOVA (assumes equal variances): "no difference"

F = 23.9, p = 0.0009
Welch's ANOVA (df = 2, 6.7), which weights each group by n_i/s_i^2: "clear difference"

When group variances differ a lot, and especially when group sizes are unequal too, use **Welch's ANOVA** (`oneway.test()` in R). Many statisticians now use it by default: when the variances really are equal, it loses almost nothing.

6.6 Checking the model: residual diagnostics

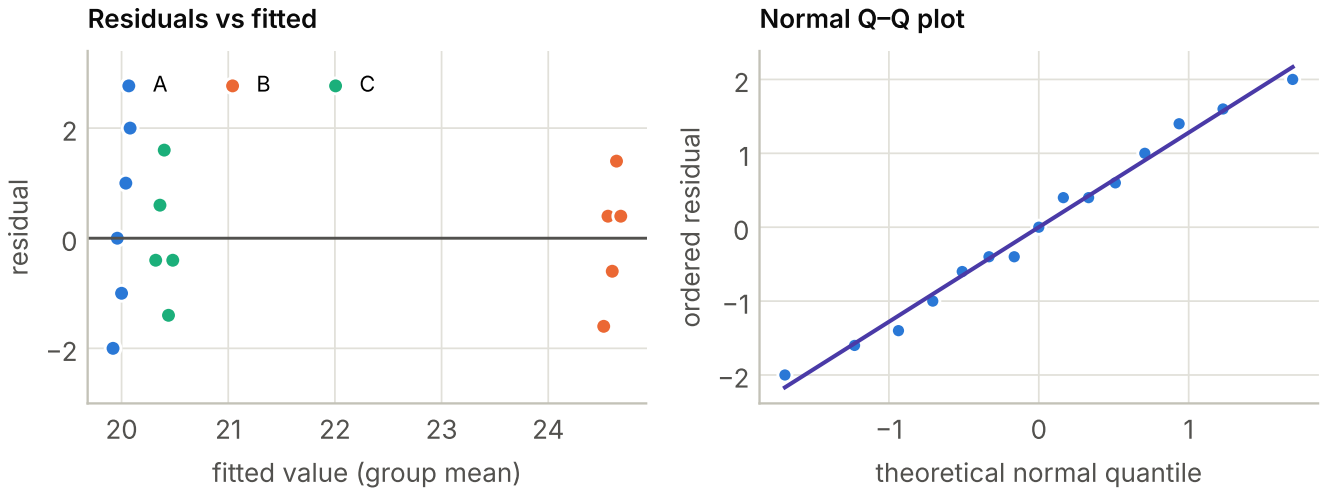


Figure 8. Diagnostics for the fertilizer model. Left: residuals vs fitted values should form an even horizontal band around zero, with no funnel (unequal variance) and no curve. Right: in a normal Q–Q plot the points should hug the line. Both look fine here. With 15 points, do not over-read small wiggles.

7 ANOVA inside regression: farm yield

Model: *yield per acre ~ farm size + fertilizer*. Both predictors are continuous, so this is multiple linear regression. But OLS regression has the same decomposition: $SST = SSR + SSE$ (model + residual). The overall regression F test *is* an ANOVA F test comparing the model with an intercept-only model.

Farm size (acres)	5	8	10	12	15	18	20	24	28	32	35	40
Fertilizer (kg)	35	45	50	55	65	70	75	85	92	100	105	115
Yield (q/acre)	21.58	21.22	22.45	22.08	24.22	24.35	25.78	25.52	27.08	27.10	28.72	29.28

$$\widehat{\text{yield}} = 18.523 + 0.119\,(\text{farm size}) + 0.0535\,(\text{fertilizer})$$

Source	SS	df	MS	F
Model	80.60	2	40.30	113.97
Residual	3.18	9	0.354	
Total	**83.78**	**11**		

Coefficient	Estimate	SE	t	p
Intercept	18.523	2.221	8.34	< 0.001
Farm size	0.119	0.176	0.68	0.52
Fertilizer	0.0535	0.0780	0.69	0.51

R^2 = SSR/SST = 0.962 (adjusted 0.954); overall p = 4.1e-07.

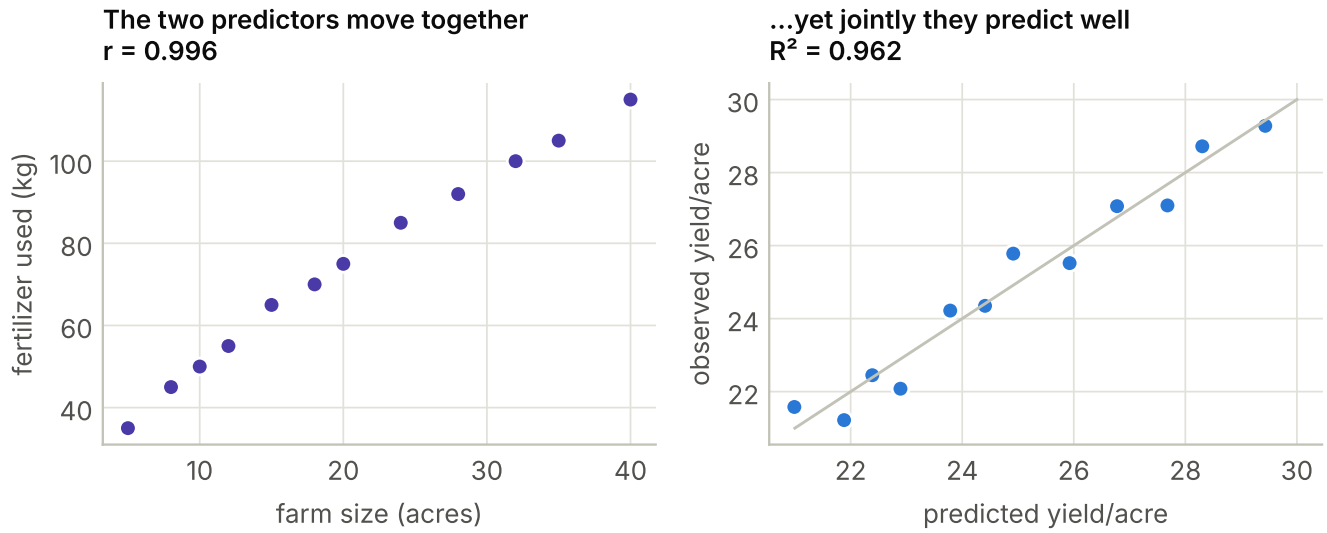


Figure 9. A puzzle, explained. Left: in these data, bigger farms almost always use more fertilizer (r = 0.996). Right: the fitted model predicts yield extremely well. The model as a whole is decisive, but it cannot tell *which* of two near-identical predictors deserves the credit.

⚠ THE COLLINEARITY TRAP (CORRECTED IN THIS EDITION)

The overall F is huge ($p \approx 4 \times 10^{-7}$), yet *neither* slope is individually significant (p ≈ 0.51 each). Alone, each predictor is highly significant ($R^2 \approx 0.96$, $p \approx 2.5 \times 10^{-8}$). The reason is **collinearity**: the variance inflation factor is $VIF = 1/(1 - r^2) \approx 123$, so each slope's variance is inflated about 123-fold. The coefficients "0.119 per acre" and "0.054 per kg" are not interpretable separately here. To separate them you would need farms where size and fertilizer vary independently, ideally by randomising fertilizer.

The previous edition printed 18.501 + 0.117·size + 0.054·fertilizer and F = 114.25. Recomputed from the printed data, the values are 18.523, 0.119, 0.0535 and F = 113.97. The original was probably fitted to unrounded yields.

7.1 A warning about causal interpretation

Regression ANOVA shows whether the predictors jointly explain variation beyond an intercept-only model. It does not, by itself, show that farm size or fertilizer *causes* yield changes. Causal claims need design: random assignment, or credible control of confounding. Each coefficient answers a partial question: the expected change in yield with one predictor, *holding the other fixed*.

8 Two-way ANOVA and interactions

With two factors, the model part of the variation can be split further: into the **main effect** of factor A, the main effect of factor B, their **interaction** A × B, and the residual. An interaction means *the effect of one factor depends on the level of the other*.

Toy design: 3 seed varieties × 3 irrigation levels × 3 replicate plots (27 plots). To keep the data fully reproducible, the three replicates in every cell are *cell mean − 1, cell mean, cell mean + 1*. So the within-cell SS is always 9 × 2 = 18 on 18 df, and $MS_{residual} = 1$. We compare two scenarios with different cell means.

Cell means	Low	Medium	High
V1	28	31	33
V2	32	34	37
V3	30	33	35

Scenario 1: no interaction

Cell means	Low	Medium	High
V1	28	31	33
V2	27	33	39
V3	33	33	32

Scenario 2: V3 is drought-tolerant; V2 is thirsty

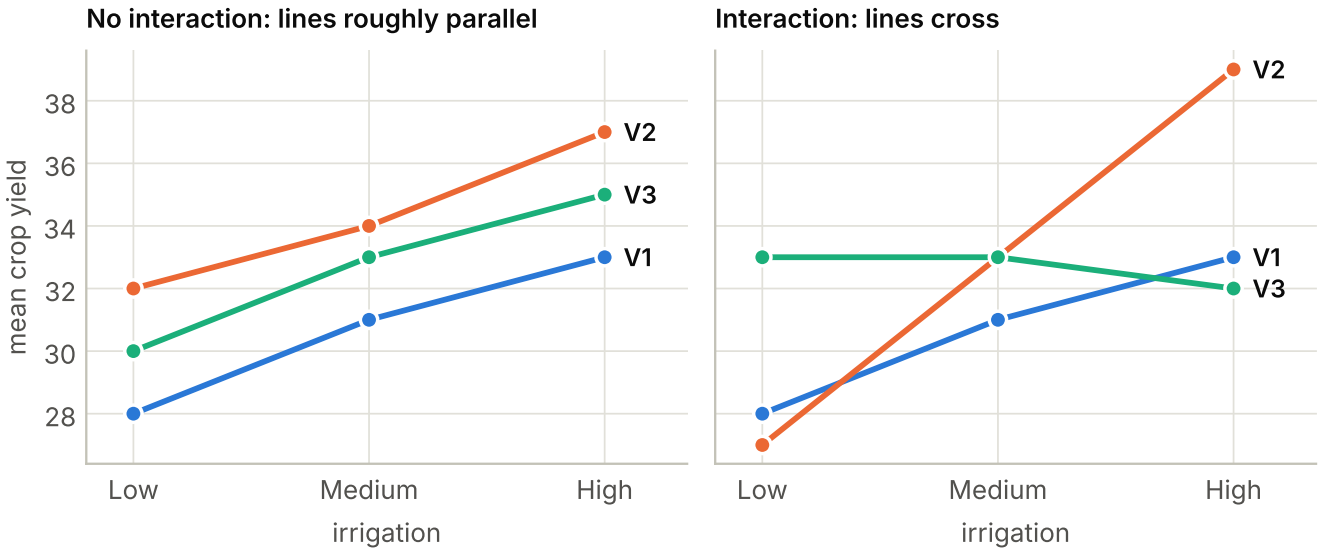


Figure 10. Interaction plots. Left: every variety gains roughly the same amount from more water, so the lines are nearly parallel (no interaction). Right: V2 responds strongly to water, V3 barely at all, and the lines cross. Asking "which variety is best?" now has no single answer: it depends on irrigation.

Source	df	Scenario 1: parallel lines			Scenario 2: crossing lines		
		SS	F	p	SS	F	p
Variety (A)	2	60.67	30.33	< 0.001	28.67	14.33	< 0.001
Irrigation (B)	2	112.67	56.33	< 0.001	128.67	64.33	< 0.001
Variety × Irrigation	4	1.33	0.33	0.852	127.33	31.83	< 0.001
Residual (within cells)	18	18.00			18.00		
Total	**26**	**192.67**			**302.67**		

$MS_{residual}$ = 18/18 = 1.00 in both scenarios, so each F equals its MS.

In Scenario 1 the interaction row is tiny (F = 0.33, p = 0.85), and the main effects can be read directly: "V2 yields about 3.7 more than V1 on average; high irrigation adds about 5 over low". In Scenario 2 the interaction is huge (F = 31.8). The variety main effect of 28.7 averages over opposite patterns and should not be interpreted on its own: *always look at the interaction first*. Notice that the residual MS is 1.00 in both scenarios, because the noise is identical. What changes is only how the *model* variation is divided up.

PART III

Surveys: clustering, ICC and DEFF

random intercepts | intraclass correlation | design effect | effective sample size | margin of error | optimal cluster size

9 Clustering, ICC and the design effect

Surveys rarely sample people independently. They sample **PSUs** (primary sampling units: villages, polling booths, census blocks, schools, households) and then interview several people inside each. People from the same PSU tend to resemble each other: they share a neighbourhood, a local candidate, the same news and the same caste mix. That resemblance means the observations are *not* independent in the way a simple random sample (SRS) of the same size would be. ANOVA is exactly the tool that measures how much they resemble each other.

9.1 The random-intercept model

$$y_{ij} = \mu + u_j + e_{ij}, \qquad u_j \sim (0, \sigma_b^2), \quad e_{ij} \sim (0, \sigma_w^2) \quad \Rightarrow \quad \mathrm{Var}(y_{ij}) = \sigma_b^2 + \sigma_w^2$$

Here u_j is cluster j's shared deviation (its "local flavour") and e_{ij} is person i's individual deviation. The **intraclass correlation** is the share of total variance that sits *between* clusters:

$$\rho = \frac{\sigma_b^2}{\sigma_b^2 + \sigma_w^2} \qquad \text{(also the correlation between two people in the same cluster)}$$

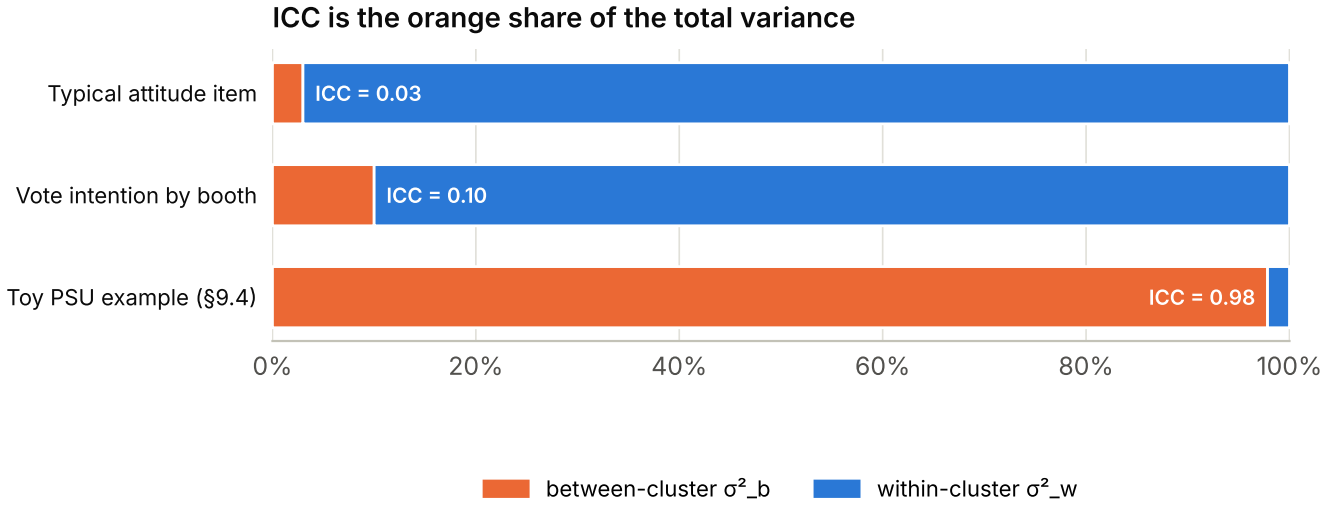


Figure 11. ICC as a share of variance. For many attitude items the between-cluster share is small (ρ ≈ 0.01–0.05). Vote intention by polling booth is often higher (ρ ≈ 0.05–0.15), because booths are socially homogeneous. The toy example in §9.4 is deliberately extreme.

9.2 Estimating the ICC from a one-way random-effects ANOVA

Treat PSU as the "group" in a one-way ANOVA. With equal cluster size m, $E[\mathrm{MS}_W] = \sigma_w^2$ and $E[\mathrm{MS}_B] = \sigma_w^2 + m\sigma_b^2$. Solving these gives the estimates:

$$\hat{\sigma}_w^2 = \mathrm{MS}_W, \qquad \hat{\sigma}_b^2 = \max\left(0,\ \frac{\mathrm{MS}_B - \mathrm{MS}_W}{m}\right), \qquad \hat{\rho} = \frac{\mathrm{MS}_B - \mathrm{MS}_W}{\mathrm{MS}_B + (m-1)\mathrm{MS}_W}.$$

The truncation at zero matters: when $\mathrm{MS_B} < \mathrm{MS_W}$ by chance, the raw estimate of σ^2_b would be negative, which is impossible for a variance.

9.3 From ICC to the design effect: a short derivation

The mean of the m people in one cluster shares the single u_j, so its variance is $\sigma_b^2 + \sigma_w^2/m$. Rewrite that using $\sigma^2 = \sigma_b^2 + \sigma_w^2$ and ρ:

$$\mathrm{Var}(\bar{y}_j) = \sigma_b^2 + \frac{\sigma_w^2}{m} = \frac{\sigma^2}{m}\left[1 + (m-1)\rho\right] \quad \Rightarrow \quad \mathrm{Var}(\bar{y}) = \underbrace{\frac{\sigma^2}{n}}_{\text{SRS}} \times \underbrace{\left[1 + (m-1)\rho\right]}_{\text{DEFF}}$$

for K independent clusters and $n = Km$. So the **design effect** is the factor by which clustering inflates the variance of the mean. The **effective sample size** is the SRS size that would give the same precision:

$$\mathrm{DEFF} = 1 + (m-1)\rho, \qquad \mathrm{DEFT} = \sqrt{\mathrm{DEFF}}, \qquad n_{\text{eff}} = \frac{n}{\mathrm{DEFF}}.$$

The formula is exact under this model (equal cluster sizes, a common ρ, clusters sampled independently). In real surveys it is an approximation. **DEFT** is what multiplies standard errors and confidence-interval widths.

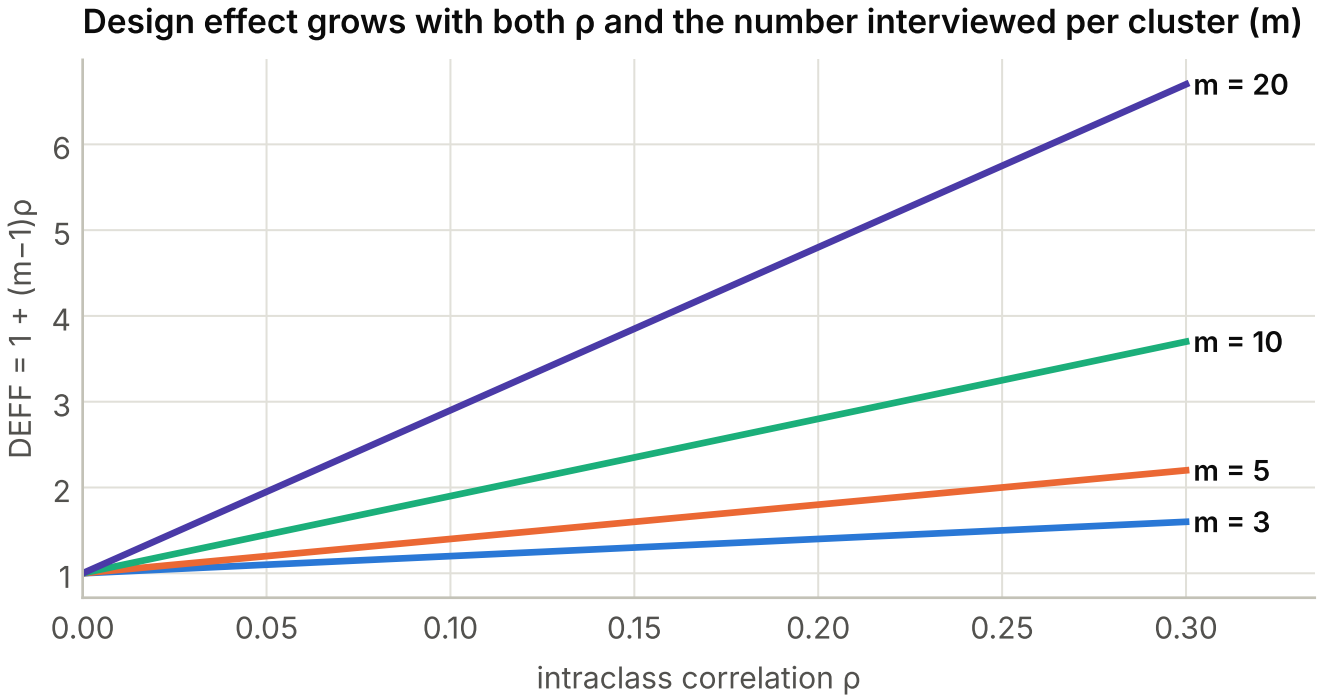


Figure 12. DEFF rises with ρ and with the number of interviews per cluster. Even a modest ρ = 0.10 doubles the variance when m = 11, and nearly triples it at m = 20. Small ICCs are not harmless when clusters are large.

9.4 Worked example: 8 PSUs × 5 respondents

A deliberately extreme toy survey (n = 40). The PSU means are 42, 47, 51, 55, 60, 64, 68 and 73, and within each PSU the five scores are mean − 2, −1, 0, +1, +2.

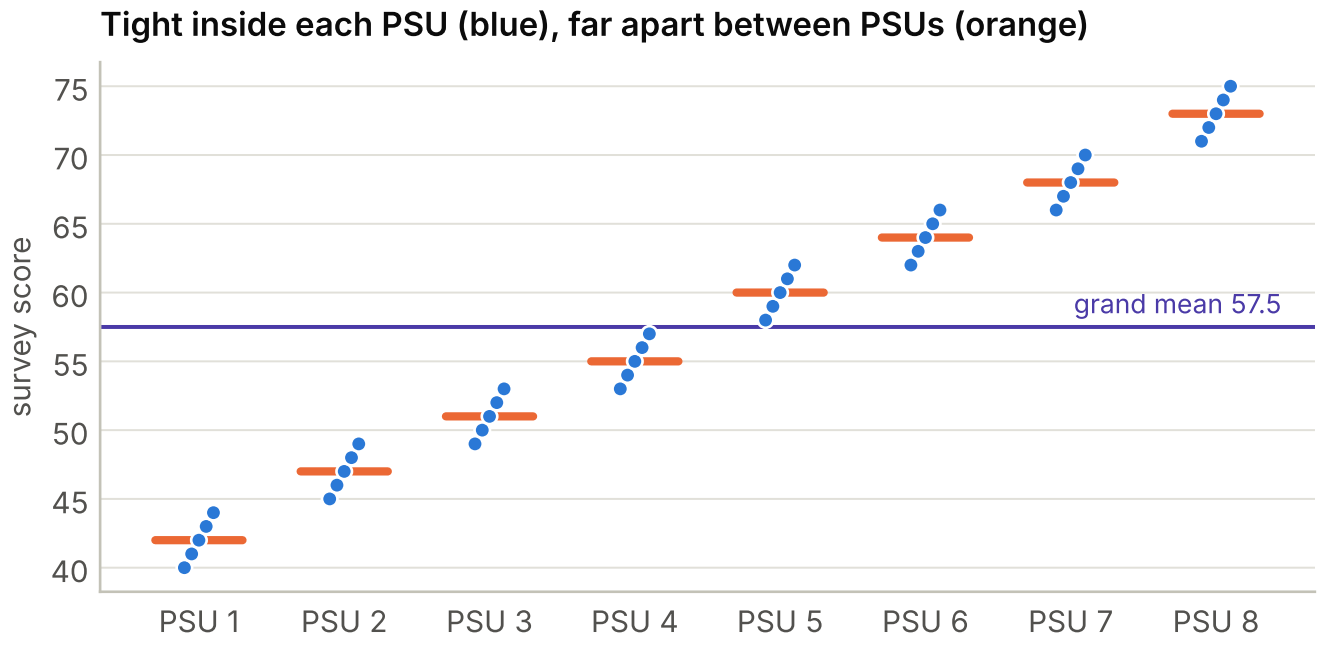


Figure 13. A strongly clustered sample. Inside each PSU the respondents are nearly identical (blue); the PSU means (orange) differ hugely. Most of the information in this sample is about *which PSU* someone lives in.

Source	SS	df	MS	F
Between PSUs	3,990.00	7	570.00	228.0
Within PSUs	80.00	32	2.50	
Total	**4,070.00**	**39**		

$\hat{\sigma}_w^2 = 2.50,\ \hat{\sigma}_b^2 = (570 - 2.5)/5 = 113.5$

$\hat{\rho} = 113.5/(113.5 + 2.5) = \mathbf{0.978}$

$\text{DEFF} = 1 + 4(0.978) = \mathbf{4.91},\ n_{\text{eff}} = 40/4.91 \approx \mathbf{8.1}$

Forty clustered interviews carry about as much information about the overall mean as eight independent ones. The maximum possible DEFF here is $m = 5$, reached when ρ = 1: then each PSU is effectively one respondent repeated five times.

9.5 Seeing it happen: a simulation

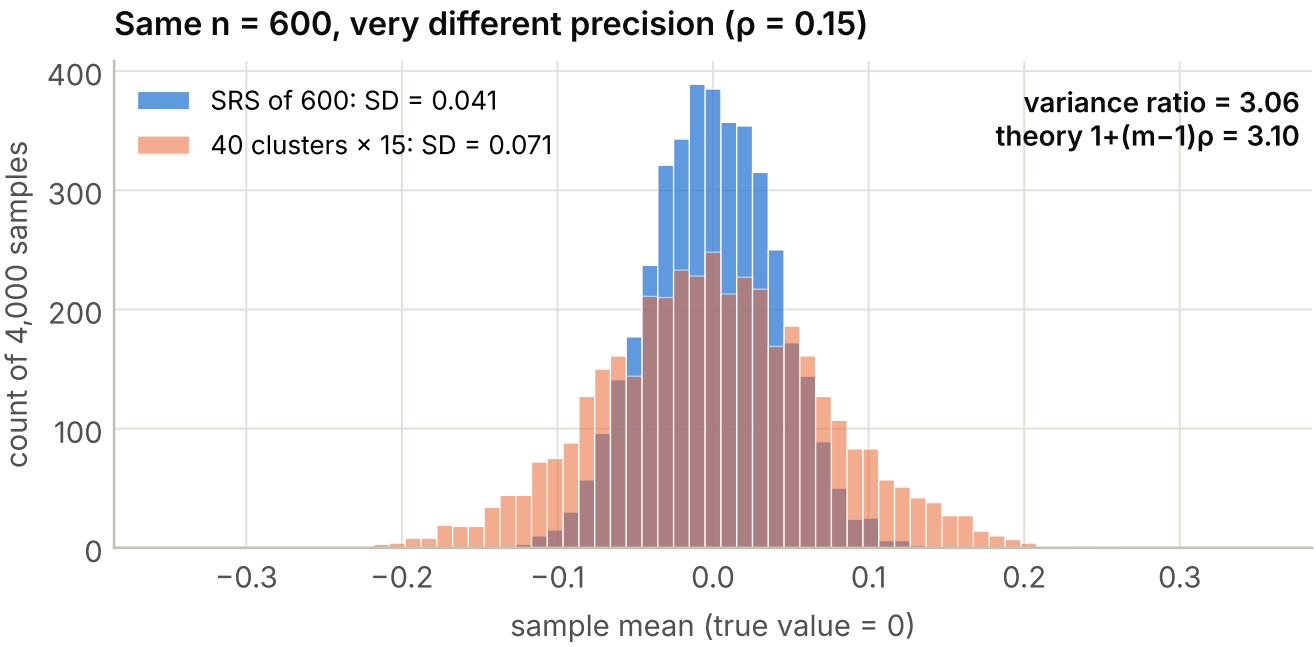


Figure 14. The design effect is not just a formula; you can watch it. We drew 4,000 samples of n = 600 from a population with ρ = 0.15: once as an SRS (blue), once as 40 clusters × 15 people (orange). Both are unbiased, centred on the truth, but the clustered estimates are much more spread out. The observed variance ratio (3.06) matches 1 + (m − 1)ρ = 3.10.

9.6 "Real" differences can still hurt precision

Between-PSU differences are genuine population heterogeneity, not measurement noise. Yet they create positive covariance among respondents who share a PSU, which inflates the sampling variance of an overall mean. DEFF does not say the between-cluster component is fake. It says each extra interview in the same PSU adds *less new information*, because part of it repeats what that PSU has already told you.

9.7 Unequal cluster sizes, weights and stratification

Unequal cluster sizes

Information concentrates in a few large clusters, raising DEFF. Kish's approximation, with CV_m the coefficient of variation of cluster sizes: $1 + \{(1 + CV_m^2)\bar{m} - 1\}\rho$

Unequal weights

Varying weights add variance. Kish's weighting effect is $1 + CV_w^2$; a rough overall DEFF is often taken as the product of the weighting and clustering parts.

Stratification

Usually *reduces* variance, when strata (region, urban/rural) are internally similar and differ from each other.

For a real survey, estimate the DEFF of each statistic directly as $\mathrm{Var}_{\mathrm{design}}(\hat{\theta}) / \mathrm{Var}_{\mathrm{SRS}}(\hat{\theta})$, using software that knows the strata, PSUs and weights (Taylor linearisation or replicate weights; the R `survey` package does this). The DEFF is *statistic-specific*: vote share, age and turnout each have their own ρ.

10 Survey lab: a Punjab-style pre-poll design

SCENARIO (ILLUSTRATIVE NUMBERS)

A pre-election survey plans **2,400 interviews** in **200 polling booths** (**m = 12** per booth). Pilot data suggest an ICC for vote intention of about **ρ = 0.06**. The headline estimate is a party's vote share of about **35%**. Post-stratification weights have $CV_w \approx 0.5$. These inputs are for teaching, not from a real survey.

10.1 What does clustering cost us?

1.66
DEFF from clustering = 1 + (12 − 1) × 0.06

1.29
DEFT: standard errors are 29% wider than under SRS

1,446
Effective sample size (of 2,400 interviews)

≈ 2.07
DEFF with weights too ≈ 1.66 × 1.25 → $n_{eff} \approx 1{,}157$

Assumed design	Standard error	95% margin of error	95% interval for 35%
Simple random sample of 2,400	0.97 pts	± 1.9 pts	33.1% – 36.9%
200 booths × 12 (clustering)	1.25 pts	± 2.5 pts	32.5% – 37.5%
Clustering + unequal weights	1.40 pts	± 2.7 pts	32.3% – 37.7%

⚠ REPORTING "± 2%" FOR A CLUSTERED SURVEY UNDERSTATES THE UNCERTAINTY

The SRS formula $1.96\sqrt{p(1-p)/n}$ gives ± 1.9 points here, but the honest design-based margin is closer to ± 2.7. The difference matters when two parties are separated by 3–4 points. Always multiply by DEFT, or better, let survey software compute the standard error from the actual design.

10.2 How many interviews per booth? The cost–precision trade-off

Opening a booth (travel, permissions, supervisor time) costs c_1; each extra interview there costs c_2. For a fixed budget, the variance of the mean is minimised by the classic result:

$$m_{\text{opt}} = \sqrt{\frac{c_1}{c_2} \cdot \frac{1-\rho}{\rho}} = \sqrt{\frac{6000}{300} \cdot \frac{0.94}{0.06}} \approx \mathbf{18} \text{ interviews per booth.}$$

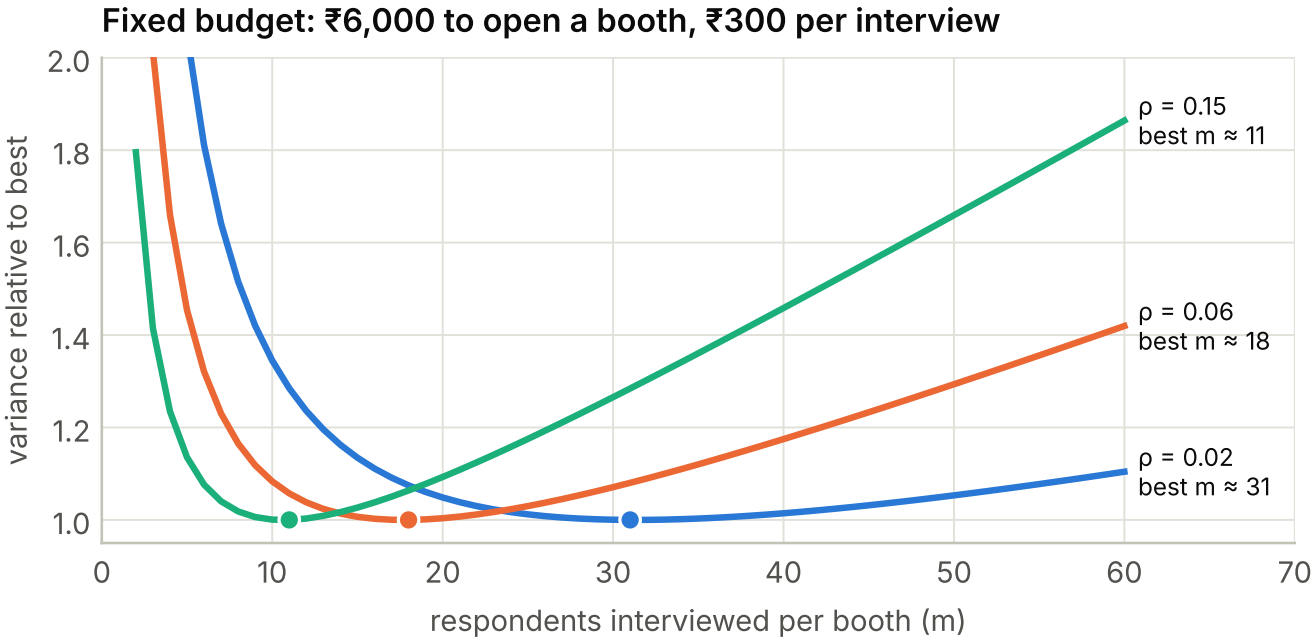


Figure 15. For a fixed budget, variance is U-shaped in m. Too few interviews per booth wastes money on travel; too many wastes interviews on people who resemble each other. The higher the ICC, the smaller the best m. Note how flat each curve is near its minimum: for ρ = 0.06, any m from about 11 to 25 is within 6% of the best possible variance.

Holding the budget at ₹1,920,000 (the cost of the planned 200 × 12 design), here is what other choices of m would buy:

Interviews per booth (m)	Booths affordable	Total interviews n	DEFF	Effective n	MOE for 35% (pts)
4	266	1,064	1.18	**902**	±3.11
8	228	1,824	1.42	**1,285**	±2.61
12	200	2,400	1.66	**1,446**	±2.46
18	168	3,024	2.02	**1,497**	±2.42
30	128	3,840	2.74	**1,401**	±2.50
50	91	4,550	3.94	**1,155**	±2.75

PRACTICAL LESSON

When ρ > 0, **more PSUs with fewer respondents each** is usually more efficient than a few PSUs sampled deeply, up to the point where field costs bite. Near the optimum the gains are small (here only a few percent over m = 12). Far from it, for example 50 interviews in each of 91 booths, the loss is large.

PART IV

Practice

assumptions | decision guide | formula sheet | R code | 32 exercises | answers

11 Assumptions, diagnostics, and a decision guide

Assumption	Why it matters	How to check	If it fails
Independence most important	Clustered, repeated or paired data make ordinary SEs too small (Part III).	Think about the design: PSUs, households, repeated visits, same interviewer.	Mixed / multilevel model, repeated-measures, cluster-robust SEs, survey design methods.
Equal variances	The pooled MS_W can hide or exaggerate differences (Example 12).	Residuals-vs-fitted funnel; compare group SDs (ratio > 2 is a warning).	Welch's ANOVA (`oneway.test`); robust (HC) SEs in regression.
Normality of residuals	Matters for small samples; ANOVA is fairly robust with balanced groups.	Q–Q plot of residuals (Fig. 8), not of the raw outcome.	Transform (log), rank-based Kruskal–Wallis, or bootstrap.
Correct model	Residual variance is "noise" only relative to the fitted model.	Residual patterns; plot against omitted variables and time.	Add covariates, interactions, curvature.
Multiplicity	Many pairwise tests inflate false alarms.	Count your comparisons.	Tukey HSD, planned contrasts, Holm correction.
Effect size	p depends on sample size as much as on the effect.	Always ask "how big?"	Report mean differences with CIs, ω^2, partial η^2, R^2, variance components.

11.1 A practical decision guide

Your question / data structure	Typical model	In R
One categorical factor, independent groups	One-way ANOVA (linear model with a factor)	`aov(y ~ g)`
...and group variances clearly differ	Welch's ANOVA	`oneway.test(y ~ g)`
Which specific groups differ?	Tukey HSD / planned contrasts	`TukeyHSD(fit)`
Two categorical factors	Two-way (factorial) ANOVA with interaction	`aov(y ~ A * B)`
Factor + continuous covariate	ANCOVA / multiple regression	`lm(y ~ g + x)`
Several continuous predictors	Multiple regression + ANOVA F tests	`anova(lm0, lm1)`
Repeated measurements on the same unit	Repeated-measures / mixed model	`lmer(y ~ t + (1|id))`
People nested in villages / booths / schools	Multilevel (random-intercept) model; ICC	`lmer(y ~ 1 + (1|psu))`
Survey estimate with weights, strata, PSUs	Design-based estimator (Taylor linearisation or replicate weights)	`svydesign()`, `svymean()`
Skewed outcome, small groups, outliers	Kruskal–Wallis or bootstrap	`kruskal.test(y ~ g)`

12 Formula sheet

TOTAL SS & VARIANCE

$$\text{SST} = \sum_i (y_i - \bar{y})^2, \quad s^2 = \text{SST}/(n-1)$$

ONE-WAY IDENTITY

$$\text{SST} = \text{SSB} + \text{SSW}$$

BETWEEN SS

$$\text{SSB} = \sum_i n_i(\bar{y}_i - \bar{y})^2, \quad df = k-1$$

WITHIN SS

$$\text{SSW} = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2, \quad df = n-k$$

F STATISTIC

$$F = \frac{\text{MS}_B}{\text{MS}_W} \sim F_{k-1,\,n-k} \text{ under } H_0$$

EXPECTED MEAN SQUARES

$$E[\text{MS}_W] = \sigma^2, \; E[\text{MS}_B] = \sigma^2 + \frac{\sum n_i(\mu_i - \bar{\mu})^2}{k-1}$$

EFFECT SIZES

$$\eta^2 = \frac{\text{SSB}}{\text{SST}}, \quad \omega^2 = \frac{\text{SSB} - (k-1)\text{MS}_W}{\text{SST} + \text{MS}_W}$$

REGRESSION

$$\text{SST} = \text{SSR} + \text{SSE}, \quad R^2 = \text{SSR}/\text{SST}$$

$$\text{VIF}_j = 1/(1 - R_j^2)$$

RANDOM INTERCEPT & ICC

$$\sigma^2 = \sigma_b^2 + \sigma_w^2, \quad \rho = \sigma_b^2/\sigma^2$$

ICC FROM ANOVA (EQUAL M)

$$\hat{\rho} = \frac{\text{MS}_B - \text{MS}_W}{\text{MS}_B + (m-1)\text{MS}_W}$$

DESIGN EFFECT

$$\text{DEFF} = 1 + (m-1)\rho$$

$$\text{DEFT} = \sqrt{\text{DEFF}}, \quad n_{\text{eff}} = n/\text{DEFF}$$

UNEQUAL CLUSTERS & WEIGHTS (KISH)

clusters: $1 + \{(1 + CV_m^2)\bar{m} - 1\}\rho$

weights: $1 + CV_w^2$

MARGIN OF ERROR FOR A PROPORTION

$$\text{MOE} = 1.96\sqrt{\text{DEFF} \cdot p(1-p)/n}$$

OPTIMAL INTERVIEWS PER CLUSTER

$$m_{\text{opt}} = \sqrt{\frac{c_1}{c_2} \cdot \frac{1-\rho}{\rho}}$$

13 Do it in R (and Python)

Everything in this chapter can be reproduced with a few lines of R. The full, tested script `ANOVA_companion.R` is saved next to this PDF; the key steps are below, with the output they produce.

13.1 One-way ANOVA from first principles, then in one line

```r
fert ← data.frame(
  type  = factor(rep(c("A", "B", "C"), each = 5)),
  yield = c(18,20,19,21,22,  23,25,24,26,25,  20,21,22,19,20))

grand  ← mean(fert$yield)
gmeans ← ave(fert$yield, fert$type)        # each row's own group mean
SSB ← sum((gmeans - grand)^2)              # signal
SSW ← sum((fert$yield - gmeans)^2)         # noise
F   ← (SSB / 2) / (SSW / 12)               # 19.1

fit ← aov(yield ~ type, data = fert)       # the same thing in one line
summary(fit)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
type         2  64.93   32.47    19.1 0.000187 ***
Residuals   12  20.40    1.70
```

13.2 After the F test: Tukey, Welch and diagnostics

```r
TukeyHSD(fit)                                  # which pairs differ?
oneway.test(hours ~ firm, data = courier)      # Welch: unequal variances
par(mfrow = c(1, 2)); plot(fit, which = 1:2)   # residual plots
```

```
     diff       lwr       upr     p adj
B-A   4.6  2.400023  6.799977 0.0003272
C-A   0.4 -1.799977  2.599977 0.8796157
C-B  -4.2 -6.399977 -2.000023 0.0007152

F = 23.907, num df = 2.0000, denom df = 6.7448, p-value = 0.0008674
```

13.3 Regression, collinearity and the two-way model

```r
m1 ← lm(yield ~ size + fert, data = farm)
anova(lm(yield ~ 1, data = farm), m1)      # the regression F IS an ANOVA
car::vif(m1)                               # ~123: severe collinearity
summary(aov(yield ~ variety * irrigation, data = cells))
```

13.4 ICC and design effect: by ANOVA, by mixed model, by survey design

```r
tab ← summary(aov(score ~ psu, data = svy))[[1]]
MSB ← tab$`Mean Sq`[1];  MSW ← tab$`Mean Sq`[2];  m ← 5
icc ← (MSB - MSW) / (MSB + (m - 1) * MSW)       # 0.978
1 + (m - 1) * icc                                # DEFF = 4.91

library(lme4)                                    # same ICC from a mixed model
vc ← as.data.frame(VarCorr(lmer(score ~ 1 + (1 | psu), data = svy)))$vcov
vc[1] / sum(vc)

library(survey)                                  # design-based SE and DEFF
des ← svydesign(ids = ~booth, weights = ~wt, data = pun)
svymean(~vote_A, des, deff = "replace")
```

```
           mean       SE   DEff
vote_A 0.365417 0.013005 1.7498
```

(Simulated data for 200 booths × 12: the survey package's DEFF of ≈ 1.75 is close to the planning value $1 + 11\rho$.)

13.5 The same in Python

```python
import statsmodels.formula.api as smf, statsmodels.api as sm
fit = smf.ols("yield_ ~ C(type)", data=fert).fit()
sm.stats.anova_lm(fit)                                  # one-way ANOVA table
from scipy.stats import f_oneway, tukey_hsd
f_oneway(a, b, c); tukey_hsd(a, b, c)                     # F test and Tukey
smf.mixedlm("score ~ 1", svy, groups=svy["psu"]).fit()  # variance components → ICC
```

14 Exercises

★ warm-up ★★ core ★★★ stretch NEW R Work with pencil first, then check in R.

★ Warm-up: concepts and small calculations

1. In your own words, define variance. Why are deviations from the mean squared?
2. For the data 4, 6, 8, compute the mean, SST and sample variance.
3. Explain the difference between total variance and within-group variance in a one-way ANOVA.
4. What does the null hypothesis of a one-way ANOVA state?
5. Why is an F statistic near 1 broadly compatible with the null hypothesis?
6. Group A is 10, 12, 11 and Group B is 15, 14, 16. Compute both group means and the grand mean.
7. Using the data in Q6, compute SSW.
8. Using the data in Q6, compute SSB and verify that SST = SSB + SSW.

★★ Core: tables, interpretation, regression and design

9. For Q6–Q8, compute $df_{between}$, df_{within}, $MS_{between}$, MS_{within} and F.

10 Explain why a statistically significant ANOVA does not tell you which particular group means differ.

11 A study has four treatments with n = 8 per treatment. What are $df_{between}$, df_{within} and df_{total}?

12 Suppose SST = 240 and SSW = 90. Compute SSB and η^2. Interpret η^2.

13 Suppose $MS_{between} = 30$ and $MS_{within} = 5$. Compute F. What is the signal-to-noise interpretation?

14 Give three reasons why residual variance may contain systematic structure rather than pure random noise.

15 Explain how randomisation and replication make the within-group variance more interpretable as noise.

16 In the model yield/acre ~ farm size + fertilizer, what does the overall regression F test ask?

17 A regression has SST = 500 and SSE = 125. Compute SSR and R^2. What proportion remains unexplained in-sample?

18 In a two-way ANOVA, distinguish a main effect from an interaction using an agricultural example.

19 A two-way model has a strong interaction. Why can interpreting a main effect alone be misleading?

20 Describe when Welch's ANOVA would be preferred to ordinary one-way ANOVA.

21 In a cluster survey, $\sigma^2_{between} = 9$ and $\sigma^2_{within} = 36$. Compute the total variance and the ICC.

22 For Q21, if m = 6 respondents per PSU, compute DEFF and n_{eff} for a nominal n = 600.

★★★ Stretch: survey design and estimation

23 Two designs both sample n = 600. Design A takes 100 PSUs × 6 people; Design B takes 30 PSUs × 20 people. If $\rho = 0.08$, compute the equal-m DEFF for each and compare their effective sample sizes.

24 A one-way random-effects ANOVA of 50 clusters of size m = 4 gives $MS_{between} = 18$ and $MS_{within} = 10$. Estimate σ^2_w, σ^2_b, the ICC and DEFF.

25 A survey has unequal cluster sizes with mean $\bar{m} = 10$, $CV_m = 0.6$ and $\rho = 0.05$. Use $DEFF \approx 1 + \{(1 + CV^2_m)\,\bar{m} - 1\}\rho$. Then explain why a full survey-design variance estimator is still preferable.

NEW Seven new exercises: pictures, pitfalls, planning and R

26 ★ Look at Figure 4. The top-left and top-right panels have the *same* group means. Without calculating, explain why one F is large and the other small.

27 ★★ Battery example (§6.4): F = 0.73, p = 0.50. A manager writes "the three brands last equally long". What is wrong with this sentence, and what would you write instead?

28 ★★ Courier example (§6.5): compute the three group variances. Explain in two sentences why classical ANOVA (p = 0.38) and Welch's ANOVA (p = 0.0009) disagree.

29 ★★ From the fertilizer ANOVA table, compute ω^2 and compare it with $\eta^2 = 0.761$. Why is ω^2 smaller, and why does the gap shrink as n grows?

30 ★★★ Punjab lab: 2,400 interviews in 200 booths, $\rho = 0.06$, $p = 0.35$. (a) Compute DEFF, DEFT and n_{eff}. (b) Compute the 95% MOE under SRS and under clustering. (c) If weights have $CV_w = 0.5$, what is the approximate overall DEFF and MOE?

31 ★★★ With $c_1 = ₹6{,}000$ per booth, $c_2 = ₹300$ per interview and $\rho = 0.06$, find m_{opt}. For a budget of ₹19,20,000, how many booths can you afford at $m = 18$? Compare n_{eff} with the 200×12 design. Is the change worth it?

32 R ★★★ In `ANOVA_companion.R`, change the within-PSU deviations from (−2, −1, 0, 1, 2) to (−10, −5, 0, 5, 10). Predict the new MS_W, σ^2_b, ICC, DEFF and n_{eff} by hand, then run the script to check.

15 Answer key and solution hints

1 Variance is the average squared distance from the mean (with a df adjustment in a sample). Squaring avoids cancellation and emphasises larger deviations.

2 Mean = 6; $SST = (4-6)^2 + 0^2 + (8-6)^2 = 8$; $s^2 = 8/2 = 4$.

3 Total variance uses deviations from the grand mean. Within variance uses deviations from each group's own mean, and estimates residual scatter after group membership is fitted.

4 All population group means are equal: $\mu_1 = \mu_2 = \ldots = \mu_k$.

5 Under H_0 both mean squares estimate the same σ^2, so their ratio fluctuates around 1.

6 Means: A = 11, B = 15, grand mean = 13.

7 A: 1 + 1 + 0 = 2; B: 0 + 1 + 1 = 2; SSW = 4.

8 $SSB = 3(11-13)^2 + 3(15-13)^2 = 24$. $SST = 28 = 24 + 4$ ✓.

9 $df_B = 1$, $df_W = 4$; MSB = 24; MSW = 1; F = 24.

10 The omnibus test evaluates the joint null. Finding which pairs differ needs contrasts or post-hoc comparisons with multiplicity control (e.g. Tukey).

11 $n = 32$; $df_B = 3$; $df_W = 28$; $df_T = 31$.

12 SSB = 150; $\eta^2 = 150/240 = 0.625$: in this sample, 62.5% of SST is associated with treatment.

13 F = 6: the between-group variation per df is six times the error mean square, far more than noise alone would typically produce.

14 Omitted covariates, nonlinear relationships, interactions, time trends, measurement error, latent subgroups, or dependence (clustering).

15 Randomisation protects against systematic confounding of treatment labels; replication shows the scatter among comparably treated units.

16 Whether the predictors, jointly, reduce the unexplained SS relative to an intercept-only model; equivalently, H_0: both slopes are zero.

17 SSR = 375; $R^2 = 0.75$; 25% of SST remains as residual in-sample.

18 Main effect: the average effect of irrigation across all varieties. Interaction: irrigation raises yield a lot for V2 but hardly at all for V3 (Fig. 10).

19 With a strong interaction, a main effect averages effects that differ greatly or even reverse across levels of the other factor.

20 When group variances are materially unequal, especially with unequal group sizes. It is a reasonable default.

21 Total variance = 45; ICC = 9/45 = 0.20.

22 DEFF = 1 + 5(0.20) = 2.0; $n_{eff} = 600/2 = 300$.

23 A: DEFF = 1 + 5(0.08) = 1.40, $n_{eff} \approx 429$. B: DEFF = 1 + 19(0.08) = 2.52, $n_{eff} \approx 238$. More clusters with fewer people each is far more efficient.

24 $\sigma^2_w = 10$; $\sigma^2_b = (18 - 10)/4 = 2$; ICC = 2/12 = 0.167 (check: 8/(18 + 3·10) = 8/48); DEFF = 1 + 3(0.167) = 1.50.

25 DEFF ≈ 1 + (1.36 × 10 − 1)(0.05) = 1 + 12.6 × 0.05 = 1.63. A design-based estimator uses the actual cluster sizes, weights, strata, finite-population corrections and the specific statistic.

26 F divides the between-group signal by within-group noise. The means (signal) are identical, but the right panel's noise is three times larger, so MS_W is about 9× larger and F shrinks about 9-fold.

27 A non-significant F is not proof of equality: with n = 5 per brand the study can only rule out differences larger than about ±0.6 hours. Better: "No evidence of a difference; differences up to ~0.6 h cannot be excluded."

28 Variances: Fast 0.7, Budget 107.7, Local 1.7. Classical ANOVA pools them into $MS_W = 36.7$, dominated by Budget, which drowns the tight Fast-vs-Local gap. Welch weights each group by n_i/s_i^2, so the precise groups count for more.

29 $\omega^2 = (64.93 - 2 \times 1.70)/(85.33 + 1.70) = 0.707$. η^2 counts the SSB that noise alone would produce (about $(k - 1)\cdot MS_W$); that expected amount stays fixed while SST grows with n, so the correction matters less in large samples.

30 (a) DEFF = 1.66, DEFT = 1.29, $n_{eff} \approx 1{,}446$. (b) SRS: ±1.9 pts; clustered: ±1.9 × 1.29 ≈ ±2.5 pts. (c) DEFF ≈ 1.66 × 1.25 ≈ 2.08; MOE ≈ ±2.7 pts; $n_{eff} \approx 1{,}157$.

31 $m_{opt} = \sqrt{(20 \times 15.67)} \approx 17.7$. At m = 18 each booth costs ₹11,400, so you can afford 168 booths (3,024 interviews); DEFF = 2.02, $n_{eff} \approx 1{,}497$ vs 1,446 for 200 × 12: only about 3.5% better. Near the optimum the curve is flat, so operational factors (supervision, interviewer fatigue) can reasonably decide.

32 Each PSU's squared deviations now sum to 250, so $MS_W = 8 \times 250/32 = 62.5$; MS_B is unchanged at 570. $\sigma^2_b = (570 - 62.5)/5 = 101.5$; ICC = 101.5/164 = 0.619; DEFF = 1 + 4(0.619) = 3.48; $n_{eff} \approx 11.5$. More within-PSU noise means *less* clustering and more information per interview.

CLOSING PERSPECTIVE

The most durable way to understand ANOVA is not as a special test for three or more means, but as an **accounting system for variation**. Total variation is split into components linked to specified structure plus a residual. Regression changes the form of the predictors but keeps the same decomposition. Multilevel models add explicit variance components. The survey design effect then follows naturally: a large between-cluster component creates intraclass correlation, which reduces the independent information each extra respondent in a PSU adds.

Once this viewpoint is clear, ANOVA, regression, ICC, random effects, cluster sampling, DEFF and effective sample size become parts of *one* coherent framework rather than a set of disconnected formulas.